

Long Answer Questions

UNIT-1

1. MO THEORY postulates
2. MO DIAGRAMS OF O₂ & CO
3. Pi MOLECULAR ORBITALS OF BENZENE
4. Pi MOLECULAR ORBITALS OF 1,3 BUTADIENE
5. Schrödinger wave equation
6. Particle in a 1 dimension box

UNIT-2

1. Types of semiconductor, applications
2. Types of Super capacitors, applications
3. Types of super conductors, properties & applications
4. Fullerene structure, properties & applications
5. CNT's types, properties & applications
6. Nanomaterials: types, properties & applications
7. Graphene: properties & Applications
8. Core shell nano materials
9. HPC Nano materials

UNIT-3

1. Nernst equation and problems
2. Conductometric titration
3. Potentiometric titration
4. Electrochemical sensors
5. Batteries
6. Fuel cells

UNIT-4

1. Conducting polymers
2. Biodegradable polymers
3. Differences between thermoplastic and thermosetting polymers
4. Mechanisms of free radical, cationic & anionic polymerisation
5. Preparation, properties and applications of Bakelite, PVC, Nylon 6,6, Buna-S, Buna-N, Carbon fiber

UNIT-5

1. Electronic transitions

2. Laws of absorption (Beer's–Lambert law)
3. UV: Principle, instrumentation, applications
4. IR: Principle & types of vibrations, Instrumentation
5. IR Applications
6. Chromatography: principle, classification & applications